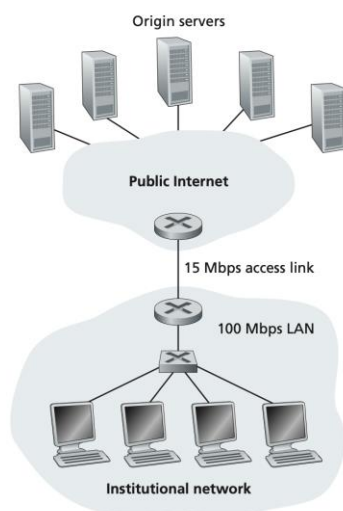


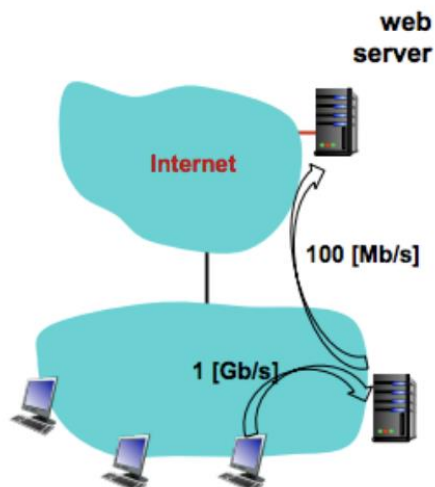
CS341 Home work Problems

1. Consider a DASH system for which there are N video versions (at N different rates and qualities) and N audio versions (at N different rates and qualities). Suppose we want to allow the player to choose at any time any of the N video versions and any of the N audio versions.
 - a. If we create files so that the audio is mixed in with the video, so server sends only one media stream at given time, how many files will the server need to store (each a different URL)?
 - b. If the server instead sends the audio and video streams separately and has the client synchronize the streams, how many files will the server need to store?
2. Consider the Figure below, for which there is an institutional network connected to the Internet. Moreover, assume the access link has been upgraded to 54 Mbps, and the institutional LAN is upgraded to 10 Gbps. Suppose that the average object size is 1,600,000 bits and that the average request rate from the institution's browsers to the origin servers is 24 requests per second. Also suppose that the amount of time it takes from when the router on the Internet side of the access link forwards an HTTP request until it receives the response is three seconds on average. Model the total average response time as the sum of the average access delay (that is, the delay from Internet router to institution router) and the average Internet delay. For the average access delay, use $\Delta/(1 - \Delta b)$, where Δ is the average time required to send an object over the access link and b is the arrival rate of objects to the access link.
 - a. Find the total average response time.
 - b. Now suppose a cache is installed in the institutional LAN. Suppose the miss rate is 0.3. Find the total response time.



3. An HTTP client requests to an HTTP server a web page constituted by one main object (an HTML file) and 10 other objects. Each object has a size equal to $L=200$ kbit. The bidirectional connection between the client and the server has a capacity of $C=100$ kbit/s. The control messages used to open the TCP connection between the client and the server, as well as the HTTP “GET” message, have a size $m=100$ bit. The propagation delay of the link is $T_p = 100$ ms. Compute the total time necessary for the client to receive the web page and the 10 objects in the two following cases:
 - a. The HTTP client opens in parallel and in a non-persistent way all TCP connections necessary to download the web page and the 10 related objects (let us assume that the transmission rate of all TCP connections is equal to $r = C/N$, where N is the number of connections open in parallel)
 - b. The HTTP client opens serially 11 TCP connections, in a non-persistent way.

4. An organization owns a Local Area Network (LAN) with an HTTP proxy having a local cache (see Figure below). Clients are connected to the HTTP proxy with a dedicated high-speed connection having capacity $C=1$ Gb/s. The probability that the content (a web page) requested by a generic client is found in the cache of the local proxy (cache hit rate) is $P=0.4$ (of course, the probability that the web page is not found in the cache is equal to $Q=0.6$). Calculate the average delay experienced by a generic client, from the instant in which it sends the HTTP request for a web page to the time in which it obtained the requested page. Let us assume that:
 - The http request messages have size of 100 byte
 - The requested webpage has size equal to 100 kbyte
 - The HTTP proxy has a communication channel with equivalent capacity $c = 100$ Mb/s towards the origin server that hosts the requested webpages
 - The opening and closing time for TCP connections between the client and the proxy as well as between the proxy and the server is negligible
 - The propagation delays are negligible.



5. Frames of 1000 bits are sent over a 1-Mbps channel using a geostationary satellite whose propagation time from the earth is 270 msec. Acknowledgements are always piggybacked onto data frames. The headers are very short. Three-bit sequence numbers are used. What is the maximum achievable channel utilization for
- a. Stop-and-wait?
 - b. Go-back-N?
 - c. Selective Repeat?
